

Building a Model to Fit Ungulate Densities

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Data Context

- Dataset 9 from the provided data sources
- Collected by wildlife biologist studying ungulate populations
 - (types of ungulates)
- Dataset was compiled to assess whether regional or seasonal factors affect ungulate concentrations more

Data Context

- 8 variables
 - ndvi -> measurement of vegetation and greenery
 - dist_water -> distance to water source (km)
 - elevation -> altitude
 - human_footprint -> index for presence of human traffic
 - snow_depth -> height of snowpack (cm)
 - region -> North, Central, South
 - season -> Summer, Winter
 - density -> individuals per km²
- Density was the response variable

ndvi	dist_water	elevation	human_footprint	snow_depth	region	season	density
0.648118	1.591527	1120.256	86.46314	21.88268	Central	Winter	11.42866
0.845263	4.546417	1394.537	55.03653	3.198445	Central	Summer	18.71176
0.315961	0.109515	442.1242	91.61587	51.85868	Central	Summer	13.44218
0.64371	0.37674	456.815	45.27318	22.28406	Central	Winter	11.67938
0.722965	3.930722	929.4815	61.25426	29.99407	South	Winter	10.86497
0.791675	1.174896	2091.093	36.12533	25.24354	Central	Summer	18.32689
0.572958	2.356021	1612.13	42.55828	35.95507	South	Summer	16.0191
0.349789	0.952007	1446.315	48.35538	34.93392	South	Winter	10.58822
0.353593	4.571554	1976.382	3.203257	35.96096	South	Summer	13.0595
0.749061	4.338675	530.6121	70.05288	8.068666	Central	Winter	11.6388
0.456452	2.210955	1445.71	3.349241	39.06346	South	Summer	15.22606
0.228683	4.937805	2460.177	31.23862	29.32446	Central	Summer	11.77973
0.607784	4.672405	1713.481	72.60878	31.04919	Central	Summer	14.16801
0.528041	2.954725	2237.499	86.00406	25.83423	North	Winter	9.740237
0.894581	0.61515	1622.496	60.68155	-2.68342	Central	Winter	13.98929
0.543274	3.008019	2316.507	7.502615	22.80426	Central	Summer	15.81859
0.829954	0.683263	2059.335	82.13664	21.3495	North	Summer	18.73812
0.531651	0.418457	1894.129	54.8993	22.23862	North	Winter	10.91977
0.325847	1.863636	2434.133	93.86586	44.4351	North	Summer	11.06044
0.56521	0.620778	1996.384	83.33886	17.4903	North	Summer	15.6126
0.720498	4.330796	2235.802	87.17695	20.41362	Central	Summer	15.04582
0.558375	4.656726	1679.783	30.54298	11.04435	Central	Winter	10.49664
0.332778	0.33299	1789.28	33.65959	30.19018	South	Winter	10.75819
0.241661	4.259929	1431.207	99.32436	31.71156	North	Winter	8.179
0.703074	4.806266	1645.636	83.12143	16.12745	Central	Summer	15.53323
0.226328	4.607389	2453.104	56.54135	38.39175	South	Summer	10.68988
0.274002	3.579417	544.7672	30.088	42.08391	Central	Winter	9.189414
0.245678	0.86791	1203.623	83.5143	34.00744	Central	Winter	9.853259

Research Importance

Intention of data was comparing whether general climate conditions or regional differences have a greater affect on the observed densities of ungulates

For time constraints, in this project did not explore the regional effect on densities.

I focused on the seasonal effects.

Research Question

From the given seasonal variables, what is the best model to explain the density of ungulates?

Next sections:

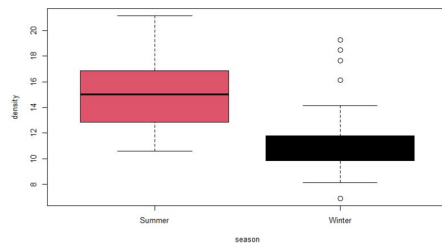
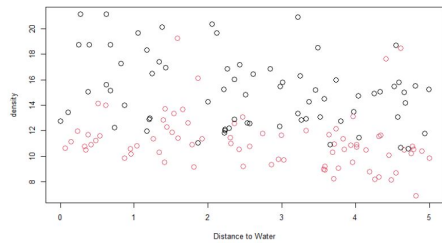
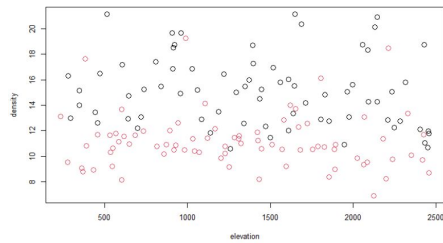
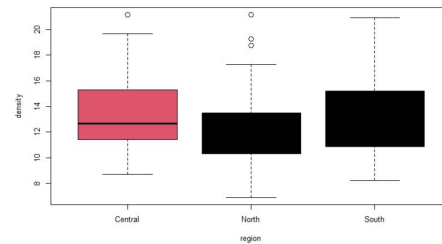
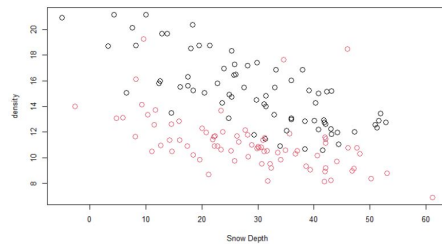
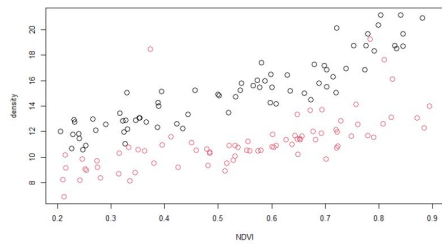
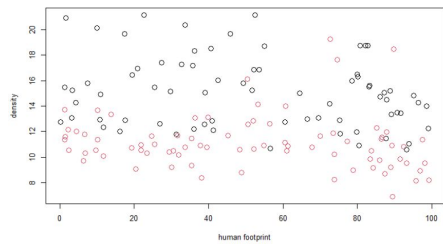
- Data exploration
- Model Building
- Diagnostics
- Model evaluation
- Final model

Data Exploration

	ndvi <dbl>	dist_water <dbl>	elevation <dbl>	human_footprint <dbl>	snow_depth <dbl>	region <chr>	season <chr>	density <dbl>
1	0.6481185	1.5915271	1120.2564	86.463142	21.882679	Central	Winter	11.42866
2	0.8452634	4.5464169	1394.5372	55.036525	3.198445	Central	Summer	18.71176
3	0.3159613	0.1095153	442.1242	91.615872	51.858680	Central	Summer	13.44218
4	0.6437102	0.3767398	456.8150	45.273182	22.284056	Central	Winter	11.67938
5	0.7229648	3.9307225	929.4815	61.254256	29.994070	South	Winter	10.86497
6	0.7916754	1.1748960	2091.0933	36.125332	25.243537	Central	Summer	18.32689
7	0.5729577	2.3560213	1612.1296	42.558284	35.955066	South	Summer	16.01910
8	0.3497887	0.9520071	1446.3149	48.355380	34.933924	South	Winter	10.58822
9	0.3535932	4.5715543	1976.3815	3.203257	35.960957	South	Summer	13.05950
10	0.7490605	4.3386752	530.6121	70.052883	8.068666	Central	Winter	11.63880

1-10 of 10 rows

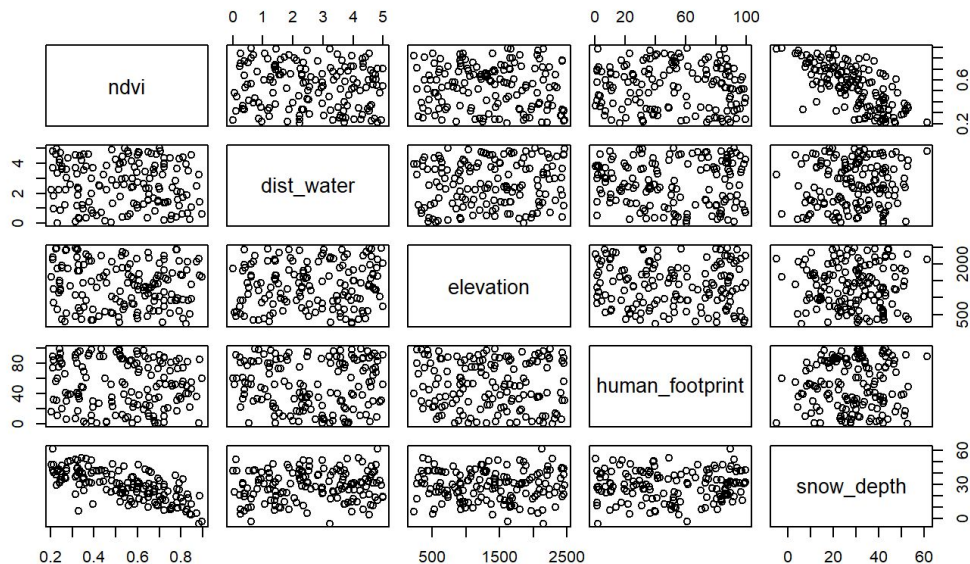
Data Exploration



Data Exploration

```
pairs(ungulates[, c("ndvi", "dist_water", "elevation", "human_footprint", "snow_depth")])
```

Collinearity check
on continuous
variables



Data Exploration

```
contVar = cor(ungulates[,c("ndvi", "dist_water", "elevation", "human_footprint", "snow_depth")])
contVar
```

```
      ndvi  dist_water  elevation human_footprint  snow_depth
ndvi      1.00000000 -0.08719734 -0.097764589   -0.05957976 -0.696893428
dist_water -0.08719734  1.00000000  0.084521535   -0.01878825  0.101914005
elevation  -0.09776459  0.08452154  1.000000000   -0.04244559 -0.003084457
human_footprint -0.05957976 -0.01878825 -0.042445590    1.00000000  0.030840547
snow_depth  -0.69689343  0.10191401 -0.003084457    0.03084055  1.000000000
```

```
##{r}
# Checking VIF, one more test of collinearity on cont. vars
full_cont_lm = lm(density ~ ndvi + dist_water + elevation + human_footprint + snow_depth,
                  data = full_ungulates)
vif_vals = vif(full_cont_lm)
VIF = as.data.frame(vif_vals)
VIF
```

	vif_vals <dbl>
ndvi	1.990099
dist_water	1.018396
elevation	1.029298
human_footprint	1.006707
snow_depth	1.973003

5 rows

snow_depth comes close, but
doesn't breach $\text{cor}() < |0.7|$

VIF < 5 not breached

Data Exploration

```
```{r}
full_ungulates$region = factor(ungulates$region,
 levels = c("Central", "North", "South"),
 labels = c("Central", "North", "South"))

full_ungulates$season = factor(ungulates$season,
 levels = c("Summer", "Winter"),
 labels = c("Summer", "Winter"))

head(full_ungulates)
```

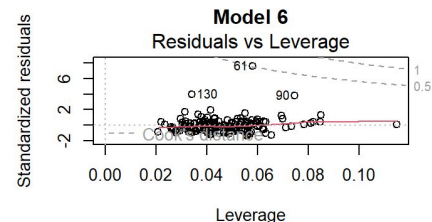
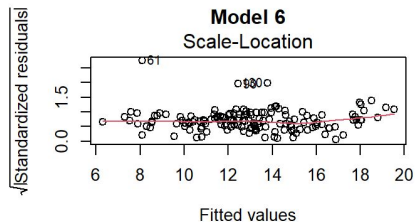
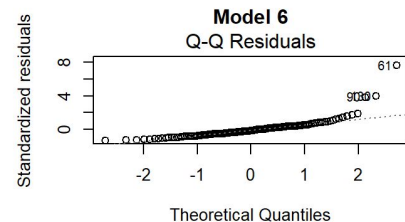
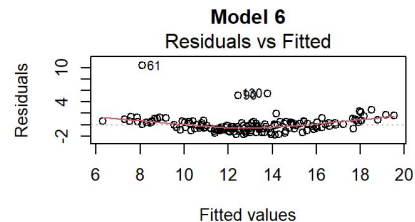
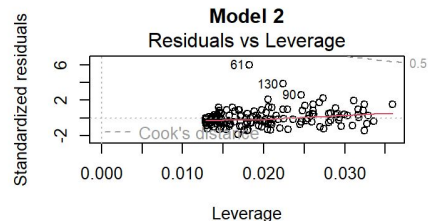
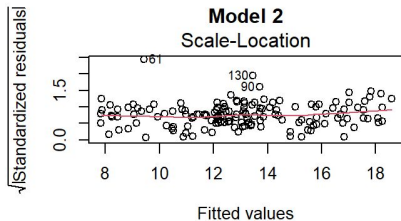
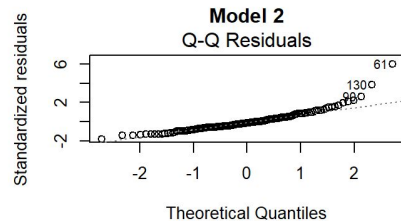
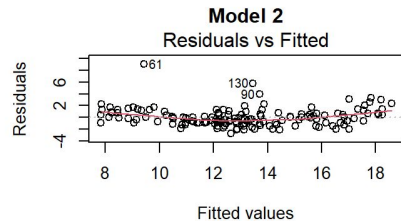
	ndvi <dbl>	dist_water <dbl>	elevation <dbl>	human_footprint <dbl>	snow_depth <dbl>	density <dbl>	region <fctr>	season <fctr>
1	0.6481185	1.5915271	1120.2564	86.46314	21.882679	11.42866	Central	Winter
2	0.8452634	4.5464169	1394.5372	55.03653	3.198445	18.71176	Central	Summer
3	0.3159613	0.1095153	442.1242	91.61587	51.858680	13.44218	Central	Summer
4	0.6437102	0.3767398	456.8150	45.27318	22.284056	11.67938	Central	Winter
5	0.7229648	3.9307225	929.4815	61.25426	29.994070	10.86497	South	Winter
6	0.7916754	1.1748960	2091.0933	36.12533	25.243537	18.32689	Central	Summer

# Model Building (Take 1)

```
non-model
model_0 = lm(density ~ 1, data = full_ungulates)

model_1 = lm(density ~ season, data = full_ungulates)
model_2 = lm(density ~ season + ndvi, data = full_ungulates)
model_3 = lm(density ~ season + ndvi + dist_water, data = full_ungulates)
model_4 = lm(density ~ season + ndvi + dist_water + elevation, data = full_ungulates)
model_5 = lm(density ~ season + ndvi + dist_water + elevation + human_footprint,
 data = full_ungulates)
model_6 = lm(density ~ season + ndvi + dist_water + elevation + human_footprint
 + snow_depth, data = full_ungulates)
...
```

# Diagnostics (Take 1)



# Diagnostics (Take 1)

Shapiro-Wilk normality test

```
data: model_2$residuals
W = 0.87148, p-value = 4.329e-10
```

Shapiro-Wilk normality test

```
data: model_3$residuals
W = 0.81706, p-value = 2.063e-12
```

Shapiro-Wilk normality test

```
data: model_4$residuals
W = 0.81406, p-value = 1.589e-12
```

Shapiro-Wilk normality test

```
data: model_5$residuals
W = 0.76103, p-value = 2.377e-14
```

Shapiro-Wilk normality test

```
data: model_6$residuals
W = 0.72108, p-value = 1.524e-15
```

Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 0.6193177, Df = 1, p = 0.4313

Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 4.360751, Df = 1, p = 0.036776

Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 4.742532, Df = 1, p = 0.029426

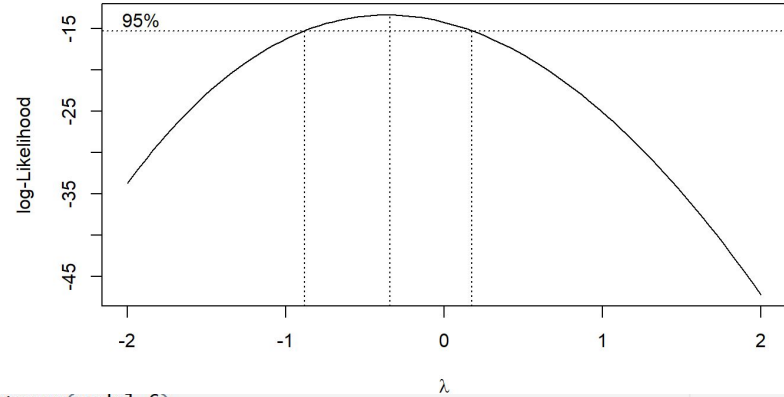
Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 12.05193, Df = 1, p = 0.00051739

Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 20.79443, Df = 1, p = 5.1131e-06

# Diagnostics (Take 1)

Based on further testing, I decided to use a  $\log()$  transformation on response

Lambda = 0 was sufficient



```
CD.model.6 <- cooks.distance(model.6)
CD.crit <- 3*mean(CD.model.6)
full_ungulates[which(CD.model.6 > CD.crit),]
```

Ultimately, removing these 4 points was required to achieve normality and homogeneity assumptions

	ndvi <dbl>	dist_water <dbl>	elevation <dbl>	human_footprint <dbl>	snow_depth <dbl>	density <dbl>	region <fctr>
39	0.8254298	1.868626	1807.2166	50.39939	8.197178	16.12204	North
61	0.3733503	4.613136	2213.6940	89.73185	45.934787	18.47730	South
90	0.8097978	4.411319	383.4791	74.64486	34.598064	17.62848	Central
130	0.7840584	1.587397	988.2534	72.74659	9.559316	19.25493	North

4 rows | 1-8 of 8 columns

# Model Building (Take 2)

```
{r}
clean_ungulates = full_ungulates[-c(39, 61, 90, 130),]
```

```
new.model.1 = lm(log(density) ~ season, data = clean_ungulates)
new.model.2 = lm(log(density) ~ season + ndvi, data = clean_ungulates)
new.model.3 = lm(log(density) ~ season + ndvi + dist_water, data = clean_ungulates)
new.model.4 = lm(log(density) ~ season + ndvi + dist_water + elevation,
 data = clean_ungulates)
new.model.5 = lm(log(density) ~ season + ndvi + dist_water + elevation
 + human_footprint, data = clean_ungulates)
new.model.6 = lm(log(density) ~ season + ndvi + dist_water + elevation
 + human_footprint + snow_depth, data = clean_ungulates)
```

First Set:

These are revised additive models, with log(density) transformation and cleaned dataset

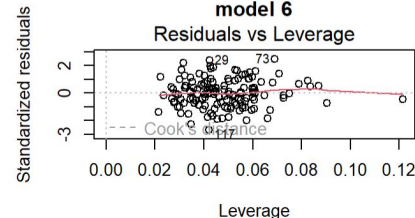
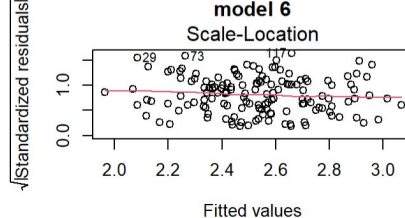
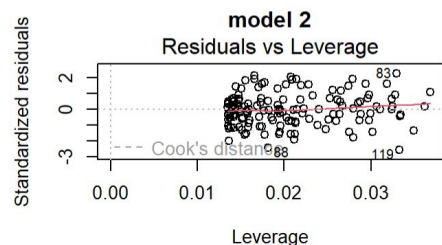
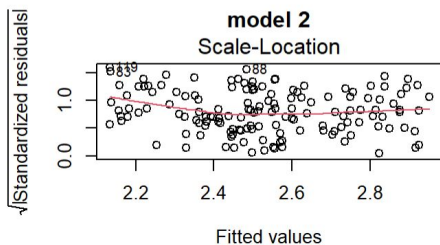
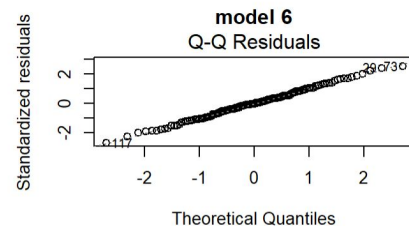
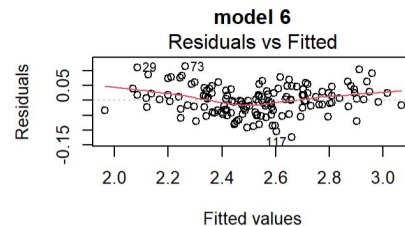
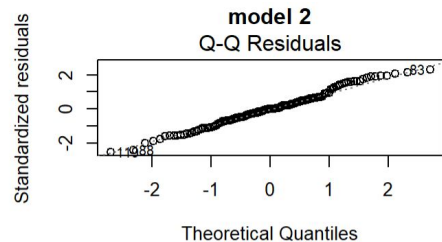
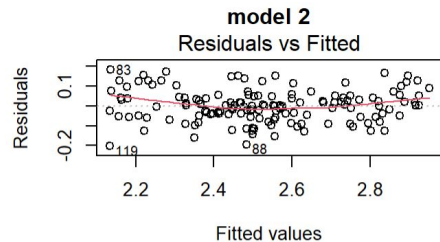
```
model.1.v3 = lm(log(density) ~ season, data = clean_ungulates)
model.2.v3 = lm(log(density) ~ season * ndvi,
 data = data = clean_ungulates)
model.3.v3 = lm(log(density) ~ season * (ndvi + dist_water),
 data = clean_ungulates)
model.4.v3 = lm(log(density) ~ season * (ndvi + dist_water + elevation),
 data = clean_ungulates)
model.5.v3 = lm(log(density) ~ season * (ndvi + dist_water + elevation
 + human_footprint),
 data = clean_ungulates)
model.6.v3 = lm(log(density) ~ season *
 (ndvi + dist_water + elevation + human_footprint + snow_depth),
 data = clean_ungulates)
```

Second Set:

Using same changes, these models include interaction between season and the other predictors

# Diagnostics (Set 1)

```
new.model.1 = lm(log(density) ~ season, data = clean_ungulates)
new.model.2 = lm(log(density) ~ season + ndvi, data = clean_ungulates)
new.model.3 = lm(log(density) ~ season + ndvi + dist_water, data = clean_ungulates)
new.model.4 = lm(log(density) ~ season + ndvi + dist_water + elevation,
 data = clean_ungulates)
new.model.5 = lm(log(density) ~ season + ndvi + dist_water + elevation
 + human_footprint, data = clean_ungulates)
new.model.6 = lm(log(density) ~ season + ndvi + dist_water + elevation
 + human_footprint + snow_depth, data = clean_ungulates)
```



# Diagnostics (Set 1)

## Shapiro-wilk normality test

data: new.model.2\$residuals  
W = 0.98846, p-value = 0.2693

Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 2.492457, Df = 1, p = 0.11439

## Shapiro-wilk normality test

data: new.model.3\$residuals  
W = 0.9931, p-value = 0.7084

Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 0.8896152, Df = 1, p = 0.34558

## Shapiro-wilk normality test

data: new.model.4\$residuals  
W = 0.99359, p-value = 0.7633

Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 0.9354059, Df = 1, p = 0.33346

```
new.model.1 = lm(log(density) ~ season, data = clean_ungulates)
new.model.2 = lm(log(density) ~ season + ndvi, data = clean_ungulates)
new.model.3 = lm(log(density) ~ season + ndvi + dist_water, data = clean_ungulates)
new.model.4 = lm(log(density) ~ season + ndvi + dist_water + elevation,
 data = clean_ungulates)
new.model.5 = lm(log(density) ~ season + ndvi + dist_water + elevation
 + human_footprint, data = clean_ungulates)
new.model.6 = lm(log(density) ~ season + ndvi + dist_water + elevation
 + human_footprint + snow_depth, data = clean_ungulates)
```

## Shapiro-wilk normality test

data: new.model.5\$residuals  
W = 0.99098, p-value = 0.4769

Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 0.1948711, Df = 1, p = 0.65889

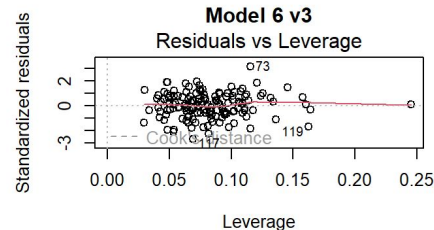
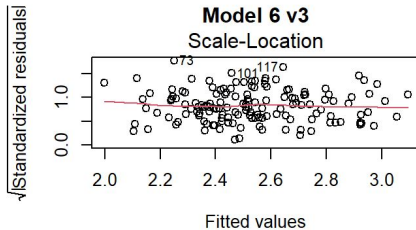
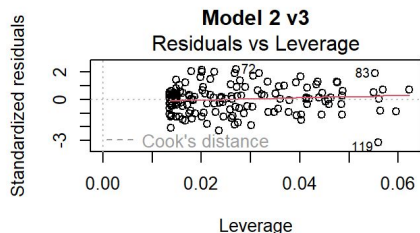
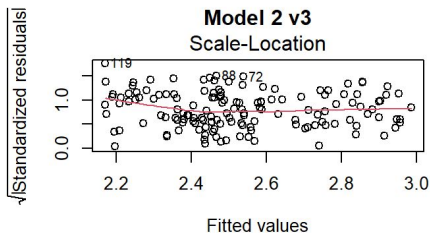
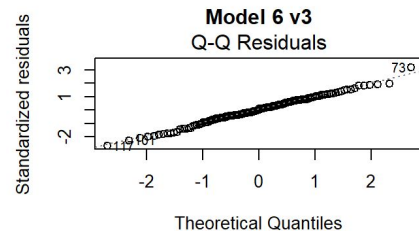
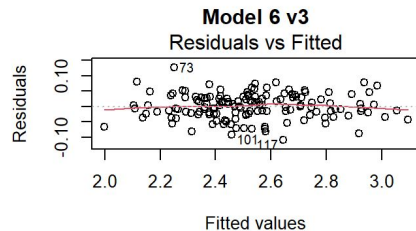
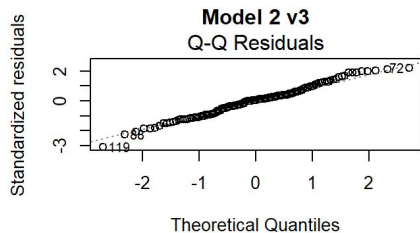
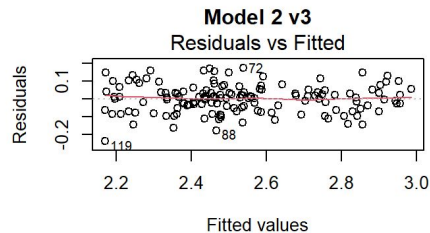
## Shapiro-wilk normality test

data: new.model.6\$residuals  
W = 0.99676, p-value = 0.9882

Non-constant Variance Score Test  
Variance formula: ~ fitted.values  
Chisquare = 0.6246597, Df = 1, p = 0.42932

# Diagnostics (Set 2)

```
model.1.v3 = lm(log(density) ~ season, data = clean_ungulates)
model.2.v3 = lm(log(density) ~ season * ndvi,
 data = data = clean_ungulates)
model.3.v3 = lm(log(density) ~ season * (ndvi + dist_water),
 data = clean_ungulates)
model.4.v3 = lm(log(density) ~ season * (ndvi + dist_water + elevation),
 data = clean_ungulates)
model.5.v3 = lm(log(density) ~ season * (ndvi + dist_water + elevation
 + human_footprint),
 data = clean_ungulates)
model.6.v3 = lm(log(density) ~ season *
 (ndvi + dist_water + elevation + human_footprint + snow_depth),
 data = clean_ungulates)
```



# Diagnostics (Set 2)

## Shapiro-wilk normality test

```
data: model.2.v3$residuals
W = 0.99041, p-value = 0.4227
```

```
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 2.738826, Df = 1, p = 0.097937
```

## Shapiro-wilk normality test

```
data: model.3.v3$residuals
W = 0.99155, p-value = 0.5361
```

```
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 1.237171, Df = 1, p = 0.26602
```

## Shapiro-wilk normality test

```
data: model.4.v3$residuals
W = 0.99218, p-value = 0.6046
```

```
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 1.249877, Df = 1, p = 0.26358
```

```
new.model.1 = lm(log(density) ~ season, data = clean_ungulates)
new.model.2 = lm(log(density) ~ season + ndvi, data = clean_ungulates)
new.model.3 = lm(log(density) ~ season + ndvi + dist_water, data = clean_ungulates)
new.model.4 = lm(log(density) ~ season + ndvi + dist_water + elevation,
 data = clean_ungulates)
new.model.5 = lm(log(density) ~ season + ndvi + dist_water + elevation
 + human_footprint, data = clean_ungulates)
new.model.6 = lm(log(density) ~ season + ndvi + dist_water + elevation
 + human_footprint + snow_depth, data = clean_ungulates)
```

## Shapiro-wilk normality test

```
data: model.5.v3$residuals
W = 0.98547, p-value = 0.1278
```

```
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 0.01313613, Df = 1, p = 0.90875
```

## Shapiro-wilk normality test

```
data: model.6.v3$residuals
W = 0.99501, p-value = 0.9002
```

```
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 0.1825325, Df = 1, p = 0.66921
```

# Model (Set 1) Estimates

```
##{r}
summary(new.model.1)
```

```
Call:
lm(formula = log(density) ~ season, data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.43914 -0.12811 0.00903 0.09788 0.35331
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.69834 0.01902 141.86 <2e-16 ***
seasonWinter -0.32902 0.02672 -12.31 <2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.1614 on 144 degrees of freedom  
Multiple R-squared: 0.5129, Adjusted R-squared: 0.5096  
F-statistic: 151.7 on 1 and 144 DF, p-value: < 2.2e-16

```
##{r}
summary(new.model.4)
```

```
Call:
lm(formula = log(density) ~ season + ndvi + dist_water + elevation,
 data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.175063 -0.044378 -0.003059 0.048446 0.148642
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.435e+00 2.392e-02 101.816 < 2e-16 ***
seasonWinter -3.385e-01 1.113e-02 -30.421 < 2e-16 ***
ndvi 6.906e-01 2.861e-02 24.137 < 2e-16 ***
dist_water -3.073e-02 3.789e-03 -8.111 2.2e-13 ***
elevation -1.452e-05 8.838e-06 -1.643 0.103

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.06674 on 141 degrees of freedom  
Multiple R-squared: 0.9185, Adjusted R-squared: 0.9162  
F-statistic: 397.1 on 4 and 141 DF, p-value: < 2.2e-16

```
##{r}
summary(new.model.2)
```

```
Call:
lm(formula = log(density) ~ season + ndvi, data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.203748 -0.052990 0.000261 0.048455 0.183322
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.32283 0.02059 112.79 <2e-16 ***
seasonWinter -0.33989 0.01347 -25.23 <2e-16 ***
ndvi 0.71381 0.03465 20.60 <2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.08131 on 143 degrees of freedom  
Multiple R-squared: 0.8772, Adjusted R-squared: 0.8755  
F-statistic: 510.9 on 2 and 143 DF, p-value: < 2.2e-16

```
##{r}
summary(new.model.5)
```

```
Call:
lm(formula = log(density) ~ season + ndvi + dist_water + elevation +
 human_footprint, data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.142231 -0.036852 0.001355 0.032789 0.110585
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.513e+00 2.163e-02 116.172 < 2e-16 ***
seasonWinter -3.406e-01 9.103e-03 -37.418 < 2e-16 ***
ndvi 6.762e-01 2.346e-02 28.827 < 2e-16 ***
dist_water -3.163e-02 3.100e-03 -10.202 < 2e-16 ***
elevation -1.754e-05 7.236e-06 -2.424 0.0166 *
human_footprint -1.226e-03 1.457e-04 -8.418 4.05e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.05457 on 140 degrees of freedom  
Multiple R-squared: 0.9459, Adjusted R-squared: 0.9439  
F-statistic: 489.2 on 5 and 140 DF, p-value: < 2.2e-16

```
##{r}
summary(new.model.3)
```

```
Call:
lm(formula = log(density) ~ season + ndvi + dist_water, data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.18761 -0.04250 -0.00234 0.05005 0.15897
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.413815 0.020279 119.032 < 2e-16 ***
seasonWinter -0.336550 0.011129 -30.240 < 2e-16 ***
ndvi 0.693899 0.028711 24.168 < 2e-16 ***
dist_water -0.031266 0.003798 -8.233 1.08e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.06713 on 142 degrees of freedom  
Multiple R-squared: 0.9169, Adjusted R-squared: 0.9151  
F-statistic: 522.3 on 3 and 142 DF, p-value: < 2.2e-16

```
##{r}
summary(new.model.6)
```

```
Call:
lm(formula = log(density) ~ season + ndvi + dist_water + elevation +
 human_footprint + snow_depth, data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.124250 -0.032431 -0.001646 0.029594 0.113980
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.675e+00 3.014e-02 88.745 < 2e-16 ***
seasonWinter -3.413e-01 7.895e-03 -43.230 < 2e-16 ***
ndvi 5.405e-01 2.837e-02 19.051 < 2e-16 ***
dist_water -3.100e-02 2.690e-03 -11.522 < 2e-16 ***
elevation -2.170e-05 6.305e-06 -3.442 0.000764 ***
human_footprint -1.256e-03 1.264e-04 -9.935 < 2e-16 ***
snow_depth -2.890e-03 4.209e-04 -6.865 2.02e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.04733 on 139 degrees of freedom  
Multiple R-squared: 0.9596, Adjusted R-squared: 0.9578  
F-statistic: 549.9 on 6 and 139 DF, p-value: < 2.2e-16

# Model (Set 1) Analysis

```
##{r}
lrttest new 3 v 4
lrttest(new.model.3, new.model.4)
```

Likelihood ratio test

```
Model 1: log(density) ~ season + ndvi + dist_water
Model 2: log(density) ~ season + ndvi + dist_water + elevation
#Df LogLik Df Chisq Pr(>Chisq)
1 5 189.22
2 6 190.60 1 2.7691 0.0961 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##{r}
lrttest new 3 v 5
lrttest(new.model.3, new.model.5)
```

Likelihood ratio test

```
Model 1: log(density) ~ season + ndvi + dist_water
Model 2: log(density) ~ season + ndvi + dist_water + elevation + human_footprint
#Df LogLik Df Chisq Pr(>Chisq)
1 5 189.22
2 7 220.50 2 62.568 2.592e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##{r}
lrttest new 5 v 6
lrttest(new.model.5, new.model.6)
```

Likelihood ratio test

```
Model 1: log(density) ~ season + ndvi + dist_water + elevation + human_footprint
Model 2: log(density) ~ season + ndvi + dist_water + elevation + human_footprint +
snow_depth
#Df LogLik Df Chisq Pr(>Chisq)
1 7 220.50
2 8 241.82 1 42.63 6.612e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	df <dbl>	AICc <dbl>
new.model.3	5	-368.0044
new.model.4	6	-368.5978

	df <dbl>	AICc <dbl>
new.model.3	5	-368.0044
new.model.5	7	-426.1891

	df <dbl>	AICc <dbl>
new.model.5	7	-426.1891
new.model.6	8	-466.5800

# Model (Set 2) Estimates

```
##{r}
summary(model.1.v3)
```

```
Call:
lm(formula = log(density) ~ season, data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.43914 -0.12811 0.00903 0.09788 0.35331
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.69834 0.01902 141.86 <2e-16 ***
seasonWinter -0.32902 0.02672 -12.31 <2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.1614 on 144 degrees of freedom  
Multiple R-squared: 0.5129, Adjusted R-squared: 0.5096  
F-statistic: 151.7 on 1 and 144 DF, p-value: < 2.2e-16

```
##{r}
summary(model.4.v3)
```

```
Call:
lm(formula = log(density) ~ season * (ndvi + dist_water + elevation),
 data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.166168 -0.042954 -0.002207 0.042997 0.138431
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.377e+00 3.138e-02 75.744 <2e-16 ***
seasonWinter -2.138e-01 4.444e-02 -4.811 3.89e-06 ***
ndvi 7.891e-01 3.752e-02 21.030 <2e-16 ***
dist_water -2.692e-02 5.286e-03 -5.094 1.13e-06 ***
elevation -1.688e-05 1.188e-05 -1.421 0.157703
seasonWinter:ndvi -2.121e-01 5.493e-02 -3.861 0.000173 ***
seasonWinter:dist_water -7.596e-03 7.269e-03 -1.045 0.297880
seasonWinter:elevation 6.315e-06 1.693e-05 0.373 0.709749

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.0639 on 138 degrees of freedom  
Multiple R-squared: 0.9268, Adjusted R-squared: 0.9231  
F-statistic: 249.7 on 7 and 138 DF, p-value: < 2.2e-16

```
##{r}
summary(model.2.v3)
```

```
Call:
lm(formula = log(density) ~ season * ndvi, data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.239875 -0.052504 0.004327 0.044685 0.172095
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.27265 0.02598 87.492 <2e-16 ***
seasonWinter -0.23041 0.03839 -6.001 1.55e-08 ***
ndvi 0.80920 0.04609 17.557 <2e-16 ***
seasonWinter:ndvi -0.20495 0.06756 -3.034 0.00287 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.07908 on 142 degrees of freedom  
Multiple R-squared: 0.8847, Adjusted R-squared: 0.8823  
F-statistic: 363.2 on 3 and 142 DF, p-value: < 2.2e-16

```
##{r}
summary(model.5.v3)
```

```
Call:
lm(formula = log(density) ~ season * (ndvi + dist_water + elevation +
 human_footprint), data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.129219 -0.026279 0.002869 0.034297 0.101393
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.442e+00 2.710e-02 90.120 <2e-16 ***
seasonWinter -1.884e-01 3.878e-02 -4.857 3.23e-06 ***
ndvi 7.815e-01 2.953e-02 26.464 <2e-16 ***
dist_water -2.753e-02 4.157e-03 -6.622 7.56e-10 ***
elevation -1.814e-05 9.343e-06 -1.941 0.0543
human_footprint -1.126e-01 1.921e-04 -5.862 3.30e-08 ***
seasonWinter:ndvi -2.279e-01 4.333e-02 -5.260 5.46e-07 ***
seasonWinter:dist_water -8.226e-03 5.719e-03 -1.438 0.1526
seasonWinter:elevation 2.437e-06 1.333e-05 0.183 0.8552
seasonWinter:human_footprint -2.386e-04 2.685e-04 -0.889 0.3758

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.05024 on 136 degrees of freedom  
Multiple R-squared: 0.9554, Adjusted R-squared: 0.9525  
F-statistic: 323.9 on 9 and 136 DF, p-value: < 2.2e-16

```
##{r}
summary(model.3.v3)
```

```
Call:
lm(formula = log(density) ~ season * (ndvi + dist_water), data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.173549 -0.039985 0.001301 0.040924 0.145895
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.351456 0.025928 90.691 <2e-16 ***
seasonWinter -0.201982 0.037830 -5.339 3.68e-07 ***
ndvi 0.794048 0.037463 21.196 <2e-16 ***
dist_water -0.027509 0.005285 -5.205 6.76e-07 ***
seasonWinter:ndvi -0.215479 0.054945 -3.922 0.000137 ***
seasonWinter:dist_water -0.007419 0.007263 -1.021 0.308816

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.06408 on 140 degrees of freedom  
Multiple R-squared: 0.9254, Adjusted R-squared: 0.9227  
F-statistic: 347.1 on 5 and 140 DF, p-value: < 2.2e-16

```
##{r}
summary(model.6.v3)
```

```
Call:
lm(formula = log(density) ~ season * (ndvi + dist_water + elevation +
 human_footprint + snow_depth), data = clean_ungulates)
```

```
Residuals:
 Min 1Q Median 3Q Max
-0.110075 -0.024035 0.001402 0.029434 0.126154
```

```
Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.615e+00 3.820e-02 68.466 <2e-16 ***
seasonWinter -2.122e-01 5.338e-02 -3.975 0.000114 ***
ndvi 6.403e-01 3.525e-02 18.166 <2e-16 ***
dist_water -2.864e-02 3.503e-03 -8.176 1.97e-13 ***
elevation -2.168e-05 7.885e-06 -2.750 0.006779 **
human_footprint -1.173e-03 1.618e-04 -7.247 3.04e-11 ***
snow_depth -3.017e-03 3.345e-04 -9.044 9.50e-08 ***
seasonWinter:ndvi -2.155e-01 5.083e-02 -4.240 4.13e-05 ***
seasonWinter:dist_water -5.090e-03 4.831e-03 -1.054 0.293926
seasonWinter:elevation 1.319e-06 1.127e-05 0.117 0.906998
seasonWinter:human_footprint -2.063e-04 2.261e-04 -0.913 0.363049
seasonWinter:snow_depth 2.830e-04 7.547e-04 0.375 0.708243

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.04226 on 134 degrees of freedom  
Multiple R-squared: 0.9689, Adjusted R-squared: 0.9664  
F-statistic: 379.8 on 11 and 134 DF, p-value: < 2.2e-16

# Model (Set 2) Analysis

```
##{r}
lrttest model 3 v 4
lrttest(model.3.v3, model.4.v3)
```

Likelihood ratio test

```
Model 1: log(density) ~ season * (ndvi + dist_water)
Model 2: log(density) ~ season * (ndvi + dist_water + elevation)
#Df LogLik Df Chisq Pr(>Chisq)
1 7 197.05
2 9 198.51 2 2.9164 0.2326
```

```
##{r}
lrttest model 3 v 5
lrttest(model.3.v3, model.5.v3)
```

Likelihood ratio test

```
Model 1: log(density) ~ season * (ndvi + dist_water)
Model 2: log(density) ~ season * (ndvi + dist_water + elevation + human_footprint)
#Df LogLik Df Chisq Pr(>Chisq)
1 7 197.05
2 11 234.69 4 75.277 1.741e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##{r}
lrttest model 5 v 6
lrttest(model.5.v3, model.6.v3)
```

Likelihood ratio test

```
Model 1: log(density) ~ season * (ndvi + dist_water + elevation + human_footprint)
Model 2: log(density) ~ season * (ndvi + dist_water + elevation + human_footprint +
snow_depth)
#Df LogLik Df Chisq Pr(>Chisq)
1 11 234.69
2 13 261.01 2 52.647 3.698e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	df <dbl>	AICc <dbl>
model.3.v3	7	-379.2902
model.4.v3	9	-377.6947

	df <dbl>	AICc <dbl>
model.3.v3	7	-379.2902
model.5.v3	11	-445.4091

	df <dbl>	AICc <dbl>
model.5.v3	11	-445.4091
model.6.v3	13	-493.2681

# Model Comparison

```
##{r}
lrtest new.model.6 v model.6.v3
lrtest(new.model.6, model.6.v3)
```

Likelihood ratio test

```
Model 1: log(density) ~ season + ndvi + dist_water + elevation + human_footprint +
 snow_depth
Model 2: log(density) ~ season * (ndvi + dist_water + elevation + human_footprint +
 snow_depth)
#Df LogLik Df Chisq Pr(>Chisq)
1 8 241.82
2 13 261.01 5 38.395 3.144e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##{r}
AICc(new.model.6, model.6.v3)
```

	df <dbl>	AICc <dbl>
new.model.6	8	-466.5800
model.6.v3	13	-493.2681

2 rows

```
##{r}
best_aicc = AICc(model.6.v3)
second_aicc = AICc(new.model.6)

ER = exp(-(second_aicc - best_aicc) / 2)
```

```
[1] 624088.2
```

After comparing the lrtest results, the AICc comparison, and looking at the adjusted  $R^2$  from the summaries:

The interaction model between seasons and all non-region predictors was the best model for explaining ungulate densities.

# Final Model

	Estimate <dbl>	Std. Error <dbl>	t value <dbl>	Pr(> t ) <dbl>
(Intercept)	2.622044e+00	2.802476e-02	93.561702	1.013810e-126
dist_water	-3.121688e-02	2.385380e-03	-13.086751	5.971765e-26
elevation	-2.102015e-05	5.590754e-06	-3.759805	2.503171e-04
human_footprint	-1.276957e-03	1.121229e-04	-11.388903	1.344139e-21
ndvi	6.432322e-01	3.007416e-02	21.388201	1.187425e-45
seasonWinter	-2.217951e-01	2.041602e-02	-10.863777	2.994895e-20
snow_depth	-2.913791e-03	3.731685e-04	-7.808245	1.308584e-12
ndvi:seasonWinter	-2.235919e-01	3.587702e-02	-6.232177	5.230637e-09

8 rows

	adjR2 <dbl>	AICc <dbl>
final model	0.9668527	-500.5144

The final best model shown here comes from dredge(), when given the full interaction model. The only interaction estimate is ndvi. The final model is would use  $y = \exp(\text{all predictors})$ , to backtransform. The only difference in the model for summer vs winter is the slope estimate for ndvi.